

Xinglin Hu

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EDUCATION

The Chinese University of Hong Kong (CUHK), Shenzhen

Sep 2023-Jun 2027

- BS in Data Science and Big Data Technology
- Cumulative GPA: **3.751/4.0** Major GPA: **3.823/4.0**

Stanford University-International Honors Program (IHP)

Jun 2025-Aug 2025

- Coursework: Machine Learning, Introduction to Experimental and Behavioral Economics

PUBLICATION & IN-PROGRESS WORK (* indicates co-first author)

[1] Sample Transform Cost-Based Training-Free Hallucination Detector for Large Language Models.

Z. Ding*, **X. Hu***, Jicong Fan.

Accepted to **NeurIPS 2026 (Poster)**.

[2] Certified Evaluation for LLMs in Optimization Modeling: From Graph Isomorphism to Formulation Isomorphism

Z. Wang, Z. Zhu, Z. Li, C. Chen, Z. Lin, M. Yang, Y. Han, Y. Lin, A. Gu, **X. Hu**, R. Sun, T. Ding

Accepted by **ICML 2026 - CTB Workshop**.

[3] ORGEval: Graph-Theoretic Evaluation of LLMs in Optimization Modeling.

Z. Wang, Z. Zhu, Z. Li, C. Chen, Z. Lin, M. Yang, Y. Han, Y. Lin, A. Gu, **X. Hu**, R. Sun, T. Ding

Submitted to **Transactions on Machine Learning Research**

PREPRINTS

[1] Abstract-LoRA: Unlocking Single-Image Style Transfer through Targeted UNet Block Training

Xinglin Hu.

arXiv preprint: arXiv:2609.13239

RESEARCH

Distribution Consistency-Based Hallucination Detection for Large Language Models

Nov 2025-Present

Project Lead, supervised by Assistant Prof. Jicong Fan, CUHK

- Proposed a novel training-free hallucination detection framework by representing prompt-conditioned LLM responses as empirical distributions over their generated-token intermediate representations
- Computed the optimal-transport (OT) distances between the token embedding sets of pairwise samples to build a Wasserstein distance matrix, which provides a measure for quantifying sample-to-sample transform costs
- Optimized OT computational costs for long-form generation by implementing a fixed shared random projection to reduce embedding dimensions to 128 while preserving core signals
- Derived two complementary detectors based on the matrix: *AvgWD* (cost magnitude) and *EigenWD* (cost-structure complexity), which jointly define a distribution-consistency detector for hallucination
- Extended the methodology to black-box LLMs via teacher forcing with 3 accessible teacher models, outperforming 5 training-free baselines on 2 closed-source LLMs across 3 challenging datasets.

Abstract-LoRA: Single-Image Style Transfer via Diffusion Models

Sep 2025-Present

Independent Research

- Proposed Abstract-LoRA, a novel framework that boosts single-image style transfer through lightweight LoRA training on specific UNet blocks in diffusion models
- Performed a fine-grained analysis of U-Net architecture using prompt injection experiments and CLIP semantic validation, successfully mapping Transformer block B1 to background features, B2 to foreground subjects, and B3 to style
- Injected LoRA weights at B1, B2, and B3 blocks, respectively, to resolve the “background loss” issue in the baseline method (B-LoRA, ECCV ’24), optimizing training efficiency by restricting LoRA to attention weights (W/Q/K/V/O)
- Implemented an abstract representation strategy using K-Means clustering and HSV-based color refinement to extract salient style features, significantly boosting style fidelity and color vibrancy with minimal content interference
- Evaluated Abstract-LoRA across diverse scenarios, achieving the highest combined content-style fidelity score (VGG+DINOv2) compared to baseline methods

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Lane Perception Model Optimization for Autonomous Driving

Jan 2026-May 2026

Undergraduate RA, Joint Research Initiative by Assistant Prof. Li Jiang (CUHK) and Huawei IAS BU

- Led the development of a multi-stage inference post-processing pipeline for the DETR-based LaneSegNet model using PyTorch and mmdetection3d, significantly reducing prediction redundancy on the OpenLane-V2 dataset
- Introduced the Non-Maximum Suppression (NMS) technique, leveraging BEV polygon IoU, centerline orientation, and topological degree weights, to resolve overlapping predictions at complex intersections
- Developed filtering modules (length, isolated nodes, IoMin) and refined topological adjacency matrices to ensure structural rationality of lane predictions
- Designed a Teacher-Student distillation framework to compress LaneSegNet, leveraging teacher predictor outputs for layer-wise supervision to reduce real-time inference latency
- Conducted ablation studies and hyperparameter tuning, visualizing BEV results via *Matplotlib* and validating pipeline effectiveness by introducing complementary evaluation metrics regarding filter performance and area coverage

Graph-Theoretic Evaluation of LLMs in Optimization Modeling

Sep 2025-Jun 2026

Undergraduate RA, supervised by Associate Prof. Ruoyu Sun, CUHK

- Assisted in developing ORGEval, a novel graph-theoretic evaluation framework that assesses LLM optimization modeling capabilities by reducing mathematical model equivalence to bipartite graph isomorphism testing
- Collaborated on coding a Weisfeiler-Lehman(WL)-test algorithm, coupled with a mathematically proven sufficient condition called “Symmetric Decomposable (SD)” to guarantee the WL test’s 100% accuracy in detecting isomorphism
- Led the construction of Bench4Opt, the first benchmark dataset comprising 394 model-data separated LP/MILP instances to evaluate LLMs’ mathematical modeling capabilities
- Benchmarked LLMs to empirically validate ORGEval’s efficiency, cutting evaluation time for hard MIPLIB instances from over an hour to 3.83 seconds, and enabling evaluation of previously unsolved open instances in just 32 seconds

INTERNSHIP

Huawei

AI & Data Analytics Intern

Jun 2026-Sep 2026

- Built an MCP query execution layer with tool schema design, SQL validation, and result formatting; Co-designed a 3-layer Skill architecture, deploying EDA, anomaly detection, and revenue analysis Skills to automate daily reporting
- Designed a main/sub-agent architecture for eCPM attribution using multiplicative decomposition, dimensional drill-down, and leave-one-out checks, isolating central decision-making from sub-agent contexts to mitigate context explosion and LLM hallucinations
- Implemented a two-tier memory system combining FIFO conversation windows with persistent query-result storage, compressing query results via statistical extraction and LLM summaries to reduce context size and enable cross-turn reuse
- Streamlined the metric knowledge base by identifying obsolete dimensions through SQL usage logs; Designed crawler-based probing with binary search to detect queryable date boundaries and classified query failures to support pre-query validation
- Developed high-registration-intent audience segments for tower-defense games by filtering and aggregating users’ registration and activation data from the warehouse, supporting targeted advertising and marketing operations

SELECTED PROJECTS

Kaggle Featured Competition: Jigsaw Agile Community Rules Classification

Oct 2025

Team Lead, Silver Medal Winner

- Implemented a Weighted Rank Ensemble approach integrating three complementary models: Qwen3-4B for in-context reasoning, GTE for contrastive learning-based semantic similarity, and DeBERTa for discriminative sequence classification
- Proposed a pseudo-labeling data augmentation strategy combining positive/negative samples pairwise, quadrupling the training data, and improving the model’s OOD reasoning capabilities
- Optimized distributed training on dual GPUs by implementing QLoRA (4-bit quantization), comprehensive LoRA fine-tuning (Attention & FFN layers), and DeepSpeed ZeRO-2

Research Seminar: Post-Training & Optimization Algorithms for Reasoning Models

Sep 2025-Dec 2025

Supervisor: Assistant Prof. Xiao Li, CUHK

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- Conducted architectural analysis of LLM reinforcement learning post-training paradigms, focusing on 7 core algorithmic limitations of standard GRPO regarding sparse rewards, gradient collapse, entropy collapse, etc.
- Evaluated 5 advanced GRPO variants (e.g., Dr GRPO, Step-GRPO, DAPO, XRPO, Off-Policy GRPO), synthesizing underlying methods (e.g., step-level dense rewards, asymmetric clipping), their theoretical underpinnings, and improvements
- Assessed second-order optimizers (Muon, Shampoo) against Adam, analyzing compute-memory trade-offs via Kronecker factorization and Fisher information approximations

HONORS & AWARDS

Silver Medal , Kaggle Featured Competition (Top 10%, 47/2445)	Oct 2025
Dean's Honours List (Twice)	Sep 2025, Sep 2024
Bowen Scholarship , CUHK-Shenzhen (Top 10%)	Sep 2023
Bronze Medal , World Mathematical Olympiad (WMO)-National Level	Dec 2016

SKILLS

- **Technical & Tools:** Python, C++, SQL, CUDA, Triton, OpenMP, OpenACC, VS Code, Adobe PS/PR, Procreate
- **Interests:** Piano (Level-10 Certification), Guitar, Painting, Basketball
- **Leadership Activities:** **Committee Member**, Harmonia College Board Game Club, CUHK-Shenzhen (2023-2025)